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Question 1

[13.5 Marks | 30 Minutes]

(a) What is k-Nearest Neighbors (k-NN)? Briefly explain the steps of **k-NN classification**. **[4 marks]**

(b) Explain how the value of ·Ĵ affects the classification result. Discuss the effects of small and large ·Ĵ . **[3.5 marks]**

(c) Describe **k-NN regression** and explain how the value of an unknown Ĵ is predicted. **[3 marks]**

(d) Write two advantages and two disadvantages of k-NN. **[3 marks]**

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Question 2

[13.5 Marks | 30 Minutes]

(a) What is a distance measure? Explain **Euclidean distance** and **Manhattan distance (L1 norm)**. **[4 marks]**

(b) Briefly describe **Mahalanobis distance** and explain when it is useful. **[3 marks]**

(c) Differentiate between **Cosine Similarity, Pearson Correlation, and Jaccard Similarity**. **[4 marks]**

(d) What is the **curse of dimensionality** in k-NN? **[2.5 marks]**

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Question 3

[13.5 Marks | 30 Minutes]

(a) Define **Linear Regression** and explain the concept of a **best-fit line**. **[3.5 marks]**

(b) Explain the role of J_{γ} and $J_{h\theta}$ in the linear regression model:

$$\dot{L} \; \urcorner \; \Sigma^{J}{}_{\gamma} R^{J}{}_{h\theta}]$$

[3 marks]

(c) Describe the **least-squares approach** for finding the best-fit regression line. **[4 marks]**

(d) Mention two applications of linear regression and explain briefly how regression can be used for prediction. **[3 marks]**

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Question 4

[13.5 Marks | 30 Minutes]

(a) What is **Polynomial Regression**? How does it differ from simple linear regression? **[3.5 marks]**

(b) Explain the concept of an **Additive Linear Model** and the use of predetermined functions f_1, f_2 . **[3.5 marks]**

(c) How can a periodic dataset be modeled using an additive model? Explain with a suitable example using a trigonometric function. **[3 marks]**

(d) Briefly describe **Gradient Descent** and **Maximum Likelihood Estimation** as alternative approaches for solving linear regression problems. **[3.5 marks]**

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Question 5

[13.5 Marks | 30 Minutes]

(a) Define **Overfitting** and **Generalization** in machine learning. **[3 marks]**

(b) Explain how increasing model complexity can affect training error and prediction accuracy. **[3.5 marks]**

(c) Why can a high-order polynomial have very low or zero training error but poor performance on unknown data? **[3.5 marks]**

(d) What is the purpose of dividing data into **training and test sets**? Mention two important considerations when making the division. **[3.5 marks]**

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Question 6

[13.5 Marks | 30 Minutes]

(a) What is **Cross-Validation**? Why is it used in machine learning? **[3 marks]**

(b) Describe the procedure of **k-fold Cross-Validation**. **[4 marks]**

(c) Explain two advantages and one disadvantage of k-fold Cross-Validation. **[3.5 marks]**

(d) What is **LOOCV (Leave-One-Out Cross-Validation)**? How is it related to k-fold Cross-Validation? **[3 marks]**

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Question 7

[13.5 Marks | 30 Minutes]

(a) Define **Probability** and explain any three basic properties of probability. **[4 marks]**

(b) What is **Conditional Probability**? Explain its meaning with a suitable example. **[3 marks]**

(c) Explain **Bayes' Rule** and describe the relationship between prior probability and posterior probability. **[4 marks]**

(d) Briefly explain why Bayesian reasoning is important in Machine Learning. **[2.5 marks]**

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Question 8

[13.5 Marks | 30 Minutes]

(a) Define **independence of events**. Write the mathematical condition for two events to be independent. **[3.5 marks]**

(b) Explain the difference between **independent events** and **mutually exclusive events**. **[3 marks]**

(c) What is **Gaussian (Normal) Distribution**? Briefly explain the role of **mean and covariance** in describing a Gaussian distribution. **[3.5 marks]**

(d) What is **Mahalanobis Distance**? Explain its relationship with data distribution and covariance. **[3.5 marks]**

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Question 9 — k-NN Classification and Effect of A

[13.5 Marks | 30 Minutes]

Consider the following two-dimensional training data:

Point	X_1	X_2	**Class**
A	1	1	Red
B	2	2	Red
C	2	4	Blue
D	4	3	Blue
E	5	5	Blue
F	6	2	Red

An unknown point $q = (3, 3)$ is to be classified using **Euclidean distance**.

(a) Calculate the Euclidean distance between q and every training point. **[4 marks]**

(b) Determine the predicted class of q for:

- $k = 1$
- $k = 3$
- $k = 5$

Show the selected neighbors and voting process in each case. **[4.5 marks]**

(c) Analyze how changing k affects the classification result in this example. Explain the relationship between **small k, variance, stability, and noise sensitivity**. **[3 marks]**

(d) If the training data are highly imbalanced toward the Blue class, explain why ordinary majority voting may produce a biased prediction and suggest a suitable modification. **[2 marks]**

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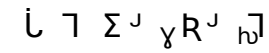
Question 10 — Simple Linear Regression from Data

[13.5 Marks | 30 Minutes]

Consider the following dataset:



Assume a linear model:



(a) Construct the required equations for determining  and using the least-squares/best-fit principle. **[4 marks]**

(b) Determine the values of and and hence obtain the best-fit regression equation. **[4 marks]**

(c) Use the resulting model to estimate when . **[2 marks]**

(d) Explain analytically why minimizing the sum of squared errors provides a criterion for finding the "best fit" line. **[3.5 marks]**

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Question 11— Overfitting and Model Selection

[13.5 Marks | 30 Minutes]

A machine-learning engineer obtains the following results:

Model	Training Error	Test Error
Linear	12	15
3rd-order Polynomial	6	8
9th-order Polynomial	0	25

(a) Identify which model is most likely **underfitting**, which provides the best generalization, and which is suffering from overfitting. Justify your answer using the training and test errors. **[4 marks]**

(b) Explain why the 9th-order polynomial can have zero training error but still perform poorly on unseen data. **[3 marks]**

(c) Analyze the relationship between increasing polynomial order, model complexity, training accuracy, and prediction accuracy. **[3 marks]**

(d) If the objective is to predict unknown data rather than merely reproduce the training data, explain which model should be selected and why. **[3.5 marks]**

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Question 12 — Training/Test Split vs. k-Fold Cross-Validation

[13.5 Marks | 30 Minutes]

A dataset contains **1,000 observations**. A researcher randomly divides it into **70% training data and 30% test data** to compare three different machine-learning models.

(a) Explain the complete procedure for selecting the best model using the training set and test set. Why should the two sets not overlap? **[3 marks]**

(b) Identify **two disadvantages** of relying on only one random training/test split. **[3 marks]**

(c) Design a **5-fold cross-validation** procedure for the same 1,000 observations. Clearly explain how many observations are used for training and testing in each fold. **[4 marks]**

(d) Explain why ·Ĉ-fold cross-validation generally provides a less split-dependent estimate of model performance than a single random train/test split. **[3.5 marks]**

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Question 13 — Conditional Probability and Bayesian Reasoning

[13.5 Marks | 30 Minutes]

A disease affects **1 person in every 1,000 people**. A screening test has the following characteristics:

- Probability of having the disease:

$$P(D) = 0.001$$

- Probability of not having the disease:

$$P(\neg D) = 0.999$$

- Probability of a positive test when the disease is present:

$$P(T^+ \mid D) = 0.99$$

- Probability of a positive test when the disease is absent:

$$P(T^+ \mid \neg D) = 0.05$$

(a) Calculate the overall probability that a randomly selected person receives a positive test result, $P(T^+)$. **[3 marks]**

(b) Using Bayes' rule, calculate:

$$P(D \mid T^+)$$

[4 marks]

(c) Interpret the result in practical terms. Why does a highly accurate positive test not imply that every person receiving a positive result actually has the disease? **[3 marks]**

(d) Clearly distinguish between **prior probability**, **likelihood/conditional probability**, and **posterior probability** in this problem. **[3.5 marks]**

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Question 14 : k-Fold Cross-Validation in Practice

(a) You have a dataset of 100 samples. You want to evaluate the performance of three different machine learning models (A, B, and C) and choose the best one. You decide to use 10-fold cross-validation. Explain the process of 10-fold cross-validation step-by-step. (6 marks)

(b) Explain why k-fold cross-validation is a better model evaluation technique than a simple single split into a training set (e.g., 70%) and a test set (e.g., 30%). Address the concepts of data waste and the statistical variance of the performance estimate. (7.5 marks)

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Question 15: Diagnosing Overfitting and Underfitting

(a) Define **overfitting** and **underfitting** in the context of machine learning models. Provide a clear example of each. (4 marks)

(b) You have plotted the training error and validation error for a model (e.g., a decision tree or polynomial regression) as the model's complexity increases. Sketch this plot. On the x-axis, show increasing model complexity, and on the y-axis, show error. Draw two curves: one for the training error and one for the validation error. Clearly label the regions of overfitting and underfitting on your graph. (5 marks)

(c) Based on your sketch in part (b), how does the gap between the training error and the validation error help diagnose overfitting? If the model is overfitting, what is the most common approach to improve its performance? (4.5 marks)